

PROOF

Constitution Propagation — PROOF (temp-dir gate run)

Note on filenames. The staged carrier drafts in this directory are stored with a `.staged` suffix (e.g. `AGENTS.md.staged`). The constitution’s propagation gates discover carriers by exact filename (`CLAUDE.md`, `AGENTS.md`, `QWEN.md`, `GEMINI.md`), so drafts named that way inside the repo are counted as real carriers and fail the gate. Drop the `.staged` suffix when you copy a file into its target submodule.

Field	Value
Revision	1
Created	2026-08-27
Last modified	2026-08-27
Status	active
Status summary	Empirical before/after for the corrected staging in this directory, measured by running all 17 CM-COVENANT-114-* -PROPAGATION gates against temp directories that mirror the real fleet. Includes a control run on the uncorrected Revision-1 drafts, so the result is not vacuous. Nothing was applied; no submodule working tree was written to; no mutating git command was run.

Continuation

Field	Value
	Operator reads this before executing <code>APPLY.md</code> §4.
§11.4.65 disclosure. This document ships without its <code>.html</code> and <code>.pdf</code> siblings — the same known, disclosed deviation <code>INVENTORY.md</code> records as gap <code>G8</code> and <code>RISKS.md</code> R7 records for this whole directory.	

1. What was being tested, and against what

`GATE-TRIAGE.md` §6.1 found that the Revision-1 staging in this directory would make the constitution gates **strictly worse**: applying `APPLY.md` §4.6 as written would take the fleet from 5 MISSING carriers / 85 MISSING lines to 8 / 136. This document is the measurement that (a) reproduces that finding, (b) shows the corrected Revision-2 staging does the opposite, and (c) proves the harness can tell the two apart.

The mechanism under test is one predicate, sourced by all 17 gates from `submodules/constitution/scripts/gates/lib/pointer_carrier.sh`:

```
BEGIN { fenced = 0; found = 0 }
/^(```|~~~)/ { fenced = !fenced; next }
!fenced && /^## INHERITED FROM / { found = 1; exit }
END { exit !found }
```

A carrier that satisfies it is scored `POINTER-INHERITANCE-SKIP` (§11.4.35 pointer consumer — legitimately omits every anchor literal). A carrier that does not, and that also lacks the anchor literal, is scored `MISSING`.

The Revision-1 drafts opened with a `>` blockquote (`AGENTS.md`, `QWEN.md`, `GEMINI.md`, the `QWEN.insert-block.md`) or with `# CLAUDE.md - <module> + ## Helix Constitution inheritance (CLAUDE.md)`. None of those is a line-anchored, non-fenced `## INHERITED FROM` heading, so every one of them registered as a **new anchor-less carrier**.

2. The first thing checked — does the claimed resolution actually hold?

Before rewriting 19 files, one candidate file was built and run through the real predicate and two real gates. The claim being tested: *the gate requires the heading to be present, non-fenced, at column zero — it does not require the path written in that heading to resolve from the file's own depth.*

```
$ . submodules/constitution/scripts/gates/lib/pointer_carrier.sh
$ is_pointer_carrier <candidate with '## INHERITED FROM
constitution/CLAUDE.md'>
PREDICATE: TRUE (pointer carrier)
$ is_pointer_carrier docs/constitution-adoption/propagation/
ai_interviewing/CLAUDE.md.staged
STAGED-CONTROL: FALSE (as triaged)
```

and through the gates themselves, on a temp dir holding only that one file:

```
$ bash submodules/constitution/scripts/gates/
cm_covenant_114_162_propagation.sh --root <tmp>
» POINTER-INHERITANCE-SKIP CLAUDE.md — §11.4.35 pointer consumer
(inherits 11.4.162 by pointer)
CM-COVENANT-114-162-PROPAGATION: 0 PRESENT, 1 POINTER-INHERITANCE-
SKIP, 0 MISSING (anchor 11.4.162)
    CM-COVENANT-114-162-PROPAGATION: PASS
rc=0
```

```
$ bash submodules/constitution/scripts/gates/
cm_covenant_114_230_propagation.sh --root <tmp>
» POINTER-INHERITANCE-SKIP CLAUDE.md — §11.4.35 pointer consumer
(engine-rules-only mirror; zero 11.4.230 blocks expected, not a
violation)
CM-COVENANT-114-230-PROPAGATION: 0 single-block-PRESENT, 1
POINTER-INHERITANCE-SKIP, 0 MISSING/DUPLICATED, 0 DIVERGENT
    CM-COVENANT-114-230-PROPAGATION: PASS
rc=0
```

Both gate families — the 12 legacy bare-literal gates and the 5 fence-aware §11.4.227(B) block-integrity gates — skip the file. **The resolution holds.** Only then were the remaining files rewritten.

3. The harness

Five temp roots, each a directory tree of CLAUDE.md / AGENTS.md / QWEN.md / GEMINI.md files at the relative paths the real repository has them at. Nothing else is needed: the gates discover carriers by filename under --root and read nothing else.

Root	Contents
baseline	byte-copies of all 20 carriers the real repository currently exposes to the gates (4 root PRESENT, 11 constitution-side pointer carriers, 5 MISSING)
defective-46	baseline + APPLY.md §4.6 applied with the Revision-1 files
defective-full	baseline + APPLY.md §4.2 – §4.6 applied with the Revision-1 files
corrected-46	baseline + APPLY.md §4.6 applied with the Revision-2 files
corrected-full	baseline + APPLY.md §4.2 – §4.6 applied with the Revision-2 files

defective-46 and defective-full are the **control**: the same harness, the same gates, the uncorrected files. If the harness could not tell them from the corrected set, the proof would be worth nothing.

The carrier list was produced with the gates' own find expression, and the §4.6 insertion used APPLY.md's own recipe verbatim ({ head -n 1 QWEN.md; echo; cat "\$B"; tail -n +2 QWEN.md; }).

Then, for each root:

```
for a in 162 167 176 187 191 196 199 200 201 202 207 213 230 231
    232 233 235; do
    bash submodules/constitution/scripts/gates/cm_covenant_114_${a}
        _propagation.sh --root <root>
done
```

3.1 Control needle — the harness reproduces reality exactly

The same 17 gates were first run against the **live repository root**, read-only:

```
$ grep -c ' MISSING' <live-run>
85
$ grep ' MISSING' <live-run> | sed 's/.*MISSING *///; s/ *_.*/'
| sort -u
milosvasic.ru/Upstreamable/AGENTS.md
milosvasic.ru/Upstreamable/CLAUDE.md
.specify/extensions/superspec/examples/static-landing-page/
CLAUDE.md
submodules/superspec/examples/static-landing-page/CLAUDE.md
vasic.digital/QWEN.md
```

CM-COVENANT-114-162-PROPAGATION: 4 PRESENT, 11 POINTER-INHERITANCE-SKIP, 5 MISSING, 17/17 gates FAIL. The baseline temp root reproduces those numbers exactly — 85 MISSING lines, the same 5 carriers, 17 FAIL — so a zero elsewhere in this document is a real zero, not a blind instrument (§11.4.201(7)(b)).

The predicate library’s own golden test was also run, unmodified:
 pointer_carrier.sh selftest: PASS (5/5), including the mid-line-backtick and ~~~~fence decoys.

4. Results

Root	gates PASS / FAIL	MISSING lines	distinct MISSING carriers	POINTER- INHERITANCE- SKIP lines
baseline (= live repo)	0 / 17	85	5	204
defective-46 (control)	0 / 17	136	8	204
defective-full (control)	0 / 17	408	24	204
corrected-46	0 / 17	68	4	272
corrected-full	0 / 17	68	4	544

Read the two rows that matter side by side, on the same gate:

defective-full CM-COVENANT-114-213-PROPAGATION: 4 PRESENT, 11 POINTER-INHERITANCE-SKIP, 24 MISSING (anchor 11.4.213)

CM-COVENANT-114-213-PROPAGATION: FAIL – 24 owned carrier(s) missing §11.4.213 anchor

corrected-full CM-COVENANT-114-213-PROPAGATION: 4 PRESENT, 31

POINTER-INHERITANCE-SKIP, 4 MISSING (anchor 11.4.213)
CM-COVENANT-114-213-PROPAGATION: FAIL – 4
owned carrier(s) missing §11.4.213 anchor

and on a fence-aware one:

defective-full CM-COVENANT-114-230-PROPAGATION: 4 single-block-
PRESENT, 11 POINTER-INHERITANCE-SKIP, 24 MISSING/DUPLICATED, 0
DIVERGENT
corrected-full CM-COVENANT-114-230-PROPAGATION: 4 single-block-
PRESENT, 31 POINTER-INHERITANCE-SKIP, 4 MISSING/DUPLICATED, 0
DIVERGENT

Every one of the 20 applied carriers is skipped by name in the
corrected run:

```
»» POINTER-INHERITANCE-SKIP ai_interviewing/AGENTS.md – §11.4.35  
pointer consumer ...  
»» POINTER-INHERITANCE-SKIP ai_interviewing/CLAUDE.md – ...  
»» POINTER-INHERITANCE-SKIP ai_interviewing/GEMINI.md – ...  
»» POINTER-INHERITANCE-SKIP ai_interviewing/QWEN.md – ...  
»» POINTER-INHERITANCE-SKIP design-toolkit/  
{AGENTS,CLAUDE,GEMINI,QWEN}.md – ...  
»» POINTER-INHERITANCE-SKIP milosvasic.ru/  
{AGENTS,CLAUDE,GEMINI,QWEN}.md – ...  
»» POINTER-INHERITANCE-SKIP monetization/  
{AGENTS,CLAUDE,GEMINI,QWEN}.md – ...  
»» POINTER-INHERITANCE-SKIP vasic.digital/  
{AGENTS,CLAUDE,GEMINI,QWEN}.md – ...
```

(The {...} braces are this document’s abbreviation; the gate prints one
line per file. $544 - 204 = 340 = 20 \text{ carriers} \times 17 \text{ gates}$, which is the
arithmetic check that no carrier was missed.)

4.1 The triage’s exact claim, reproduced

GATE-TRIAGE.md §6.1 said applying §4.6 as written takes the fleet **from 5
MISSING carriers to 8 and from 85 MISSING lines to 136.**

defective-46 measures exactly 136 / 8, and names the three new
failures:

vasic.digital/AGENTS.md
vasic.digital/CLAUDE.md
vasic.digital/GEMINI.md

on top of the pre-existing five. The triage was right.

4.2 What the corrected §4.6 does instead

corrected-46 leaves **68 / 4**. The three new carriers are skipped, and vasic.digital/QWEN.md — one of the original five MISSING — leaves the failing set entirely, because the inserted block gives it a real pointer heading. The remaining four are exactly the category-(c) third-party carriers:

```
milosvasic.ru/Upstreamable/AGENTS.md
milosvasic.ru/Upstreamable/CLAUDE.md
.specify/extensions/superspec/examples/static-landing-page/
CLAUDE.md
submodules/superspec/examples/static-landing-page/CLAUDE.md
```

Applying the *full* §4.2 – §4.6 adds 16 more carriers and changes nothing else: still 68 / 4. Sixteen new files, zero new failures.

4.3 Per-file predicate result, corrected vs. defective

All 20 target files, one command per file, both versions:

ai_interviewing/AGENTS.md	corrected=TRUE
defective=FALSE	
ai_interviewing/CLAUDE.md	corrected=TRUE
defective=FALSE	
ai_interviewing/QWEN.md	corrected=TRUE
defective=FALSE	
ai_interviewing/GEMINI.md	corrected=TRUE
defective=FALSE	
design-toolkit/AGENTS.md	corrected=TRUE
defective=FALSE	
design-toolkit/CLAUDE.md	corrected=TRUE
defective=FALSE	
design-toolkit/QWEN.md	corrected=TRUE
defective=FALSE	
design-toolkit/GEMINI.md	corrected=TRUE
defective=FALSE	
milosvasic.ru/AGENTS.md	corrected=TRUE
defective=FALSE	
milosvasic.ru/CLAUDE.md	corrected=TRUE
defective=FALSE	
milosvasic.ru/QWEN.md	corrected=TRUE
defective=FALSE	
milosvasic.ru/GEMINI.md	corrected=TRUE
defective=FALSE	
monetization/AGENTS.md	corrected=TRUE
defective=FALSE	
monetization/CLAUDE.md	corrected=TRUE
defective=FALSE	
monetization/QWEN.md	corrected=TRUE

defective=FALSE	
monetization/GEMINI.md	corrected=TRUE
defective=FALSE	
vasic.digital/AGENTS.md	corrected=TRUE
defective=FALSE	
vasic.digital/CLAUDE.md	corrected=TRUE
defective=FALSE	
vasic.digital/QWEN.md	corrected=TRUE
defective=FALSE	
vasic.digital/GEMINI.md	corrected=TRUE
defective=FALSE	

20 / 20 corrected TRUE, 20 / 20 defective FALSE. No partial credit anywhere.

5. vasic.digital/QWEN.md — the insertion is lossless

Produced with APPLY.md §4.6's own recipe, then checked:

```
orig lines: 55   new lines: 104
first 3 lines:
  # Vasic-Digital GitHub Pages Project
  (blank)
  ## INHERITED FROM constitution/QWEN.md
'## Project Overview' now at line 52

$ diff <(tail -n +2 QWEN.md.orig) <(tail -n +51 QWEN.md.new)
IDENTICAL – all 55 original lines preserved, pure insertion

$ is_pointer_carrier QWEN.md.new    -> TRUE
$ is_pointer_carrier QWEN.md.defective -> FALSE
```

Zero deletions, zero modifications to the existing 55 lines. The pointer heading sits under the file's existing H1 title, which is the shape of the four repository-root carriers that pass all 17 gates today and of the predicate's own golden-good fixture; `is_pointer_carrier()` scans the whole file, so the heading is found either way.

6. APPLY.md §5 checks, run against the corrected set

all 20 carriers: resolver named, no hardcoded parent path,
no dangling @import, pointer heading present

i.e. every file contains `find_constitution.sh`, none contains the umbrella's `submodules/constitution` layout string (a §11.4.28(B) violation), none opens with an `@constitution/ import` (which would silently resolve to nothing inside a submodule), and every one carries the `## INHERITED FROM` heading.

7. Honest boundary — what this proof does not show

- **It does not show the gates go green.** They do not. All 17 still exit 1, held by the four category-(c) third-party carriers (`GATE-TRIAGE.md` §3.1, §7). No change confined to this directory can clear them, and none was attempted. What is proven is the direction: $85 \rightarrow 68$, $5 \rightarrow 4$, and 16–19 new files that cost nothing.
- **It does not prove the propagation was applied.** Nothing was applied. Every number above comes from temp directories; no submodule working tree was written to and no mutating git command was run.
- **It does not prove the carriers' prose is correct**, only that the gates score them as pointer consumers. Whether the conditional wording is the right governance answer is `APPLY.md` §3's argument, not a gate result.
- **It does not model the design-toolkit two-parent effect.** `design-toolkit` is a gitlink of both the umbrella and the constitution, so after a commit + push + submodule update its four carriers would appear at a *second* path (`submodules/constitution/submodules/design-toolkit/`). Those copies would be pointer carriers too — the same files — so the direction does not change; the count of SKIP lines would grow by 4×17 . This is projected, not measured (measuring it needs the commit this task must not make).
- **It does not evaluate CM-CANONICAL-ROOT-CLARITY.** That §11.4.35 gate is *recommended* but not implemented (it appears in `scripts/gates/gate_ledger_prev_names.txt` with no gate script). Its clause 1. — “consumer's `CLAUDE.md` opens with the inheritance pointer” — would be satisfied by the corrected form and violated by the Revision-1 form, which is a second, independent reason for the change; but that is a reading of the anchor text, not a gate run.